ANALYTICAL CUMULATIVE EXAMINATION

November 10, 1990

1. ^{87}Rb β decays to ^{87}Sr with a 4.8×10^{10}y half-life, and this is of considerable value in geochronology. Show that the number of atoms of ^{87}Sr (D_{87}) now present in a rock sample is related to the initial number of ^{87}Sr atoms (D^{o}_{87}) and the number of ^{87}Rb (P_{87}) atoms currently present, by the relationship

$$D_{87} = D^{o}_{87} + D_{87}(e^{\lambda t} - 1)$$

where λ is the ^{87}Rb decay constant and t is time. (30%)

2. It is customary to discuss changes in isotope abundances in terms of a ratio such as $^{87}Sr/^{86}Sr$ and $^{87}Rb/^{86}Sr$. Why? (10%)

3. The following data were obtained as five samples isolated from a suite of granite rocks:

Sample	Weight ratio Rb/Sr	Isotope Ratio $^{87}Sr/^{86}Sr$
A	1.029	0.7383
B	6.424	0.8829
C	10.373	0.9913
D	9.066	0.9552
E	3.407	0.8016

Prepare a plot of $^{87}Sr/^{86}Sr$ against $^{87}Rb/^{86}Sr$ and determine the crystallization age of the rock and the initial $^{87}Sr/^{86}Sr$ ratio. (50%)

4. What significance can be attached to the initial $^{87}Sr/^{86}Sr$ ratio? (10%)

TABLE OF NUCLIDES

Z	El	Nuclide A	Abundance or $t_{1/2}$	Decay Mode	Δ(MeV)	Jπ	σ_a(b)
37	Rb	82	1.25 m	β⁻ 96%,EC 4%	-76.213	1+	
		82m	6.2 h	EC 74%,β⁺ 26%	-≈76.1	5−	
		83	86.2 d	EC	-78.914	5/2−	
		84	32.9 d	EC 75%,β⁺ 22%, β⁻ 3.0%	-79.752	2−	12 ss, 1 g
		84m	20.5 m	IT	-79.288	(6+)	0.40 g, 0.047 m
		85	72.17%	-	-82.159	5/2−	
		86	18.8 d	β⁻ 99+%,EC 0.005%	-82.738	2−	
		86m	1.02 m	IT	-82.182	6−	
		87	27.83% 4.8x10¹⁰ y	β⁻,no γ	-84.596	3/2−	0.12 ss, 1.0 s
		88	17.8 m	β⁻	-82.602	2−	
		89	15.2 m	β⁻	-81.717	(3/2−)	
		90	153 s	β⁻	-79.57	{1−}	
		90m	258 s	β⁻,IT	-79.46	{4−}	
		91	58 s	β⁻	-77.97	(1−)	
		92	4.52 s	β⁻,β⁻n 0.012%	-75.12		
		93	5.85 s	β⁻,β⁻n 1.3%	-72.92		
		94	2.72 s	β⁻,β⁻n 10%	-68.82		
		95	0.38 s	β⁻,β⁻n 8.4%	-66.55		
		96	0.201 s	β⁻,β⁻n 13%	-62.77 s		
		97	0.170 s	β⁻,β⁻n 27%			
		98	0.13 s	β⁻,β⁻n 13%			
		99	76 ms	[β⁻]			
38	Sr	77	9 s	β⁺,β⁺p ≤0.25%	-57.96		
		78	31 m	β⁺+EC	-65.5 s	0+	
		79	8.1 m	β⁺+EC	-65.46 s		
		79	4 m?	β⁺+EC			
		80	106 m	EC+β⁺	-70.39 s	0+	
		81	26 m	β⁺ ≈87%,EC ≈13%	-71.40	(1/2−)	
		82	25.0 d	EC,no γ	-75.999	0+	
		83	32.4 h	EC 76%,β⁺ 24%	-76.664	7/2+	
		83m	5.0 s	IT	-76.405	1/2−	
		84	0.56%		-80.641	0+	0.6 ss, 0.3 g
		85	64.8 d	EC	-81.095	9/2+	
		85m	68 m	IT 87%,EC 13%	-80.856	1/2−	0.84 ss
		86	9.8%		-84.512	0+	
		87	7.0%		-84.869	9/2+	
		87m	2.80 h	IT 99.7%,EC 0.3%	-84.480	1/2−	
		88	82.6%		-87.911	0+	0.0057 ss
		89	50.5 d	β⁻	-86.203	5/2+	0.42 ss
		90	28.8 y	β⁻,no γ	-85.935	0+	0.8 ss
		91	9.5 h	β⁻	-83.666	5/2+	
		92	2.71 h	β⁻	-82.892	0+	
		93	7.4 m	β⁻	-80.28		
		94	75 s	β⁻	-78.96	0+	
		95	24.4 s	β⁻	-75.14		
		96	1.1 s	β⁻	-73.07	0+	
		97	0.40 s	β⁻	-69.08 s		
		98	0.7 s	β⁻	-67.38 s	0+	
		99	0.6 s	β⁻,β⁻n 3%			
39	Y	81	5 m	β⁺+EC [β⁺]	-67.91 s		
		82	12 m?	β⁺ ≈95%,EC ≈5%	-72.36 s	(9/2+)	
		83	7.1 m	β⁺+EC	-72.36 s	{1/2}−	
		83	2.85 m	β⁺+EC	-73.692		
		84	39 m	β⁺+EC			
		84	4.6 s	β⁺+EC		(1+)	
		85(g)	2.7 h	β⁺ 55%,EC 45%	-77.855	{1/2}−	
		85(m)	4.9 h	β⁺ 70%,EC 30%	-77.835	{9/2}+	
		86	14.74 h	EC 66%,β⁺ 34%	-79.239	4−	
		86m	48 m	β⁺+EC			
				IT 99.31%, β⁺+EC 0.69%	-79.021	8+	
		87	80.3 h	EC 99.8%,β⁺ 0.2%	-83.007	1/2−	
		87m	13 h	IT 98%, EC ≈2%,β⁺ 0.75%	-82.626	9/2+	